A Modular Residue Framework for Orbitwise 2-Adic Valuations in Collatz Sequences: A Proved Prefix Density-One Law and Computational Evidence for the Full Orbit

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Abstract

We study the distribution of 2-adic valuations $K_j = v_2(3T_{\text{odd}}^j(n) + 1)$ along deterministic Collatz orbits.

Rigorous results. We prove that for each $k \geq 1$, the event $\{K_j = k\}$ is equivalent to the unique congruence $x_j \equiv a_k \pmod{2^{k+1}}$, where $a_k \equiv 3^{-1}(2^k - 1) \pmod{2^{k+1}}$ (Proposition 2). By induction we prove an exact prefix cylinder law: any finite valuation sequence $(k_0, \ldots, k_{r-1})$ corresponds to a unique odd residue class modulo 2^{S_r+1}, where $S_r = k_0 + \cdots + k_{r-1}$ (Theorem 5). This yields an exact negative-binomial prefix law (Corollary 7), and via Hoeffding's inequality and a Chernoff large-deviation bound, a density-one prefix discrepancy theorem: for any $\eta < 1/2$ and any fixed $C > 0$, the fraction of odd integers $n < 2^M$ for which the first $\lfloor \eta M \rfloor$ odd steps violate the geometric bound $|\hat{p}_k - 2^{-k}| \leq C \log(N)/\sqrt{N}$ tends to zero as $M \to \infty$ (Theorem 8). A reduction theorem shows that any orbital discrepancy bound $D_m(n, N) \leq C \log(N)/\sqrt{N}$ for all $m \leq 21$ implies the geometric valuation bound for all $k \leq 20$ (Theorem 10).

The open gap. The unresolved step is extending the prefix theorem from $N \leq \eta \log_2 n$ ($\eta < 1/2$) to the full deterministic orbit. This is the genuine frontier and is stated as an open conjecture.

Computational evidence. Exhaustive verification over 4,999,944 orbits ($n \leq 10^7$) with no violation and empirical P95 constant $C_{95} = 0.300$. Complete audit of all 2,048 odd residue classes modulo 2^{12}: the 1,725 classes with $N_{\text{odd}} \geq 10$ all pass at 100%; the 63 failures have $N_{\text{odd}} < 10$ (short-orbit artifact). Stability from 32-bit to 256-bit integers with P95 consistently below 0.300.

1 Introduction

The Collatz conjecture, formulated by Lothar Collatz in 1937, asserts that for every positive integer n, repeated application of $T(n) = n/2$ (even) or $(3n+1)/2$ (odd) eventually reaches 1. Verified for all $n \leq 2^{68}$, it remains one of the most celebrated open problems in mathematics.

1.1 Prior Work

Terras [1] and Everett [2] established that almost all integers have finite stopping time. Krasikov and Lagarias [4] proved the set satisfying the conjecture has lower asymptotic density at least 0.84. Tao [5] proved that almost all N (logarithmic density) have orbits falling below any $f(N) \to \infty$, introducing Syracuse random variables for $v_2(3X_j + 1)$ in a probabilistic model; his bounds apply to the random model, not to the deterministic orbit of a fixed integer. Inselmann [6] proved the liminf of average stopping times is $\geq 2/\log(4/3) \approx 6.95$. Lagarias [3] surveys the literature. The 2-adic dynamical setting is developed by Anashin-Khrennikov [8].

1.2 The Gap

To the author's knowledge, no published work establishes that

$$\left| \frac{1}{N} \#\{0 \leq j < N : K_j(n) = k\} - 2^{-k} \right| \leq \frac{C \log N}{\sqrt{N}}$$

for the *deterministic orbit* of a fixed integer n, for all $k \geq 1$, and with an explicit universal constant C.

1.3 Contributions

1. Exact residue characterization of $\{K_j = k\}$ (Proposition 2).
2. Exact prefix cylinder law by induction (Theorem 5).
3. Exact negative-binomial prefix law (Corollary 7).
4. Density-one prefix discrepancy theorem via Hoeffding-Chernoff (Theorem 8 and Theorem 9).
5. Reduction theorem: discrepancy bound implies valuation bound (Theorem 10).
6. Complete audit of all 2,048 odd residue classes modulo 2^{12} (Section 6.3).
7. Multi-scale computational evidence from $n \leq 10^7$ to 256-bit integers (Section 6).

2 Notation and Preliminaries

Notation. $v_2(m)$: 2-adic valuation. $U(x) = (3x+1)/2^{v_2(3x+1)}$: odd iterate map; $x_j := U^j(n)$. $K_j(n) := v_2(3x_j + 1)$. $\hat{p}_k(n, N) := \frac{1}{N}\#\{0 \leq j < N : K_j = k\}$. $C(n) := \max_{k \geq 1} |\hat{p}_k - 2^{-k}|/(\log N/\sqrt{N})$. $\mathbb{Z}_2$: 2-adic integers.

2.1 The Collatz Map

$T(n) = n/2$ (n even) or $(3n+1)/2$ (n odd).

Definition 1. For $m \geq 1$, the *orbital discrepancy modulo* 2^m is

$$D_m(n, N) := \max_{a \text{ odd}} \left| \frac{1}{N} \sum_{j<N} \mathbf{1}[x_j \equiv a \pmod{2^m}] - \frac{1}{2^{m-1}} \right|.$$

3 Modular Residue Framework

3.1 Single-Step Characterization

Proposition 2. *For each $k \geq 1$, there is a unique odd residue class $a_k \pmod{2^{k+1}}$ such that $v_2(3x+1) = k \iff x \equiv a_k \pmod{2^{k+1}}$, where*

$$a_k \equiv 3^{-1}(2^k - 1) \pmod{2^{k+1}}.$$

Proof. $v_2(3x+1) = k$ iff $3x \equiv 2^k - 1 \pmod{2^{k+1}}$. Since $\gcd(3, 2^{k+1}) = 1$, the unique solution is $x \equiv 3^{-1}(2^k - 1)$. Among 2^k odd classes modulo 2^{k+1}, exactly one satisfies this. □

Remark 3. Explicit values (verified computationally): $a_1 \equiv 3 \pmod 4$, $a_2 \equiv 1 \pmod 8$, $a_3 \equiv 13 \pmod{16}$, $a_4 \equiv 5 \pmod{32}$, $a_5 \equiv 53 \pmod{64}$. Note that $3^{-1} \equiv 3 \pmod 4$, $3^{-1} \equiv 3 \pmod 8$, $3^{-1} \equiv 11 \pmod{16}$, producing these values via the formula. The values differ from a naive pattern: $a_k \neq 2^{k+1} - 1$ in general, reflecting the nontrivial inversion of 3.

Corollary 4. *If $\{x_j\}$ is exactly uniform over odd classes modulo 2^m, then $\hat{p}_k = 2^{-k}$ for $k = 1, \ldots, m-1$.*

3.2 Exact Prefix Cylinder Law

Theorem 5 (Prefix Cylinder Law)**.** *Fix $r \geq 1$ and positive integers $k_0, \ldots, k_{r-1}$. Let $S_r = k_0 + \cdots + k_{r-1}$. There exists a unique odd residue class $a(k_0, \ldots, k_{r-1}) \pmod{2^{S_r+1}}$ such that an odd integer x satisfies $v_2(3U^j(x) + 1) = k_j$ for $j = 0, \ldots, r-1$ if and only if*

$$x \equiv a(k_0, \ldots, k_{r-1}) \pmod{2^{S_r+1}}.$$

Proof. By induction on r. For $r = 1$: Proposition 2 gives a unique class $a_k \pmod{2^{k_0+1}}$.

Assume the claim holds for the tail $(k_1, \ldots, k_{r-1})$: there is a unique odd class $b \pmod{2^{k_1+\cdots+k_{r-1}+1}}$ for $x_1 = U(x_0)$. Since $3x_0 + 1 = 2^{k_0} x_1$, and x_1 is fixed modulo $2^{S_r - k_0 + 1}$, the right side is fixed modulo 2^{S_r+1}. Because 3 is invertible modulo every power of 2, there is a unique solution x_0 modulo 2^{S_r+1}, completing the induction. □

Corollary 6 (Exact Counting on Dyadic Shells)**.** *If $M \geq S_r + 1$, then among the 2^{M-1} odd residues modulo 2^M, exactly 2^{M-1-S_r} have valuation prefix $(k_0, \ldots, k_{r-1})$. Hence the prefix probability over uniform odd residues modulo 2^M is $2^{-S_r} = \prod_j 2^{-k_j}$.*

Proof. By Theorem 5, the prefix class is unique modulo 2^{S_r+1}, and lifts to exactly $2^{M-(S_r+1)} = 2^{M-1-S_r}$ classes modulo 2^M. □

Corollary 7 (Exact Negative-Binomial Prefix Law)**.** *For odd residues uniformly distributed modulo 2^M, the prefix sum $S_N = K_0 + \cdots + K_{N-1}$ satisfies, for $s \le M-1$:*

$$P(S_N = s) = \binom{s-1}{N-1} 2^{-s},$$

the negative-binomial distribution of a sum of N independent geometric variables with $P(K = k) = 2^{-k}$.

Proof. By Corollary 6, the fraction of starting classes with $S_N = s$ is $2^{M-1-s} \cdot \binom{s-1}{N-1}/2^{M-1} = \binom{s-1}{N-1}2^{-s}$, the standard negative-binomial probability. □

Corollary 7 shows that the prefix process is not merely asymptotically geometric: below the overflow barrier $S_N \le M-1$, it is *exactly* geometric, with exact finite-M counting formulas.

3.3 Density-One Prefix Discrepancy Theorem

Theorem 8 (Density-One Prefix Bound)**.** *Fix η with $0 < \eta < 1/2$, and let $N_M = \lfloor \eta M \rfloor$. For every fixed $C > 0$ and $K \ge 1$, the number of odd integers $n < 2^M$ for which*

$$\max_{1 \le k \le K} |\hat{p}_k(n, N_M) - 2^{-k}| > \frac{C \log N_M}{\sqrt{N_M}}$$

is at most

$$2^{M-1}\left(2K\, e^{-2C^2(\log N_M)^2} + e^{-c_\eta M}\right),$$

where $c_\eta = \log 2 + (1-\eta)\log(1-\eta) + \eta \log \eta > 0$ for $\eta < 1/2$. In particular, the exceptional fraction tends to zero as $M \to \infty$.

Proof. Let A_M be the event $S_{N_M} \le M-1$. By Corollary 7, on A_M the first N_M valuations have the same distribution as N_M i.i.d. geometric variables with $P(K = k) = 2^{-k}$. For each fixed k, Hoeffding's inequality applied to the Bernoulli indicators $\mathbf{1}[K_j = k] \in [0, 1]$ gives

$$P\big(|\hat{p}_k - 2^{-k}| > t \text{ and } A_M\big) \le 2e^{-2N_M t^2}.$$

A union bound over $k = 1, \ldots, K$ with $t = C\log(N_M)/\sqrt{N_M}$ gives the first term $2Ke^{-2C^2(\log N_M)^2}$.

For the complement A_M^c: each K_j is geometrically distributed with moment generating function $E[e^{\lambda K}] = e^\lambda/(2-e^\lambda)$ for $\lambda < \log 2$. Setting $e^\lambda = 2(1-\eta)$ and optimising the Chernoff bound gives $P(A_M^c) \le e^{-c_\eta M}$ with $c_\eta = \log 2 + (1-\eta)\log(1-\eta) + \eta\log\eta$. For $\eta < 1/2$, $c_\eta > 0$ because the binary relative entropy $(1-\eta)\log(1-\eta) + \eta\log\eta > -\log 2$ strictly. Multiplying by 2^{M-1} (number of odd residues) completes the proof. □

Theorem 9 (Uniform Prefix Window Bound)**.** *Fix $\beta > 0$ and $0 < \eta < 1/2$. Let $N_{\min}(M) = \lceil M^\beta \rceil$, $N_{\max}(M) = \lfloor \eta M \rfloor$. For every fixed $C > 0$, the fraction of odd $n < 2^M$ for which there exists $N \in [N_{\min}, N_{\max}]$ and $k \in \{1, \ldots, 20\}$ with $|\hat{p}_k(n, N) - 2^{-k}| > C\log(N)/\sqrt{N}$ tends to zero as $M \to \infty$.*

Proof. Union-bound over the $N_{\max}$ prefix lengths and 20 bins, using Theorem 8 and the fact that $e^{-2C^2(\log N)^2}$ decreases in N for $N \geq 8$. The total exceptional fraction is at most $40N_{\max}(M)\, e^{-2C^2(\log N_{\min}(M))^2} + e^{-c_\eta M}$, which tends to zero faster than any power of M. □

3.4 Reduction Theorem

Theorem 10 (Reduction)**.** *For each $k \geq 1$, $|\hat{p}_k(n,N) - 2^{-k}| \leq D_{k+1}(n,N)$. If $D_m(n,N) \leq C\log(N)/\sqrt{N}$ for all $m \leq 21$, the geometric bound holds for all $k \leq 20$.*

Proof. By Proposition 2, $\hat{p}_k$ is the empirical frequency of class a_k modulo 2^{k+1} with theoretical weight 2^{-k}. Hence $|\hat{p}_k - 2^{-k}| \leq D_{k+1}(n,N)$. □

4 The Open Problem

Conjecture 11 (Full Orbit Geometric Bound)**.** *There exists $C > 0$ such that for almost all odd n and all $k \geq 1$, $|\hat{p}_k(n,N) - 2^{-k}| \leq C\log(N)/\sqrt{N}$, where N is the full number of odd steps before reaching 1.*

Theorems 8 and 9 establish Conjecture 11 for the prefix regime $N \leq \eta \log_2 n$ with $\eta < 1/2$. The full orbit of n has approximately $(1/|\log(3/4)|)\log n \approx 4.82\log_2 n$ odd steps, which exceeds the proven prefix range. The unresolved step is passing from the prefix regime to the full orbit without a length restriction; this is the genuine frontier.

By Theorem 10, the full conjecture follows from $D_m(n,N) \leq C\log(N)/\sqrt{N}$ for all $m \geq 1$.

5 Stopping Time Corollary

Assuming Conjecture 11, $\mathbb{E}[\log(3 \cdot 2^{-K_j})] = \log(3/4) \approx -0.288$, giving $\sigma_{\text{odd}}(n) \leq \frac{\log n}{|\log(3/4)|} + O(\sqrt{\log n})$ for almost all n, recovering Inselmann's constant $2/\log(4/3)$ [6] via a conditional alternative derivation.

6 Computational Evidence

6.1 Methodology

All computations used Python 3.11 and NumPy 1.26. The exhaustive pipeline used Welford's algorithm in chunks of 10,000, peak RAM < 380 MB. $C(n)$ is computed as $\max_{k=1}^{20} |\hat{p}_k - 2^{-k}|/(\log N/\sqrt{N})$. Computational evidence is consistent with Conjecture 11 but does not constitute a mathematical proof.

6.2 Exhaustive Verification to 10^7

6.3 Complete Audit of All Residue Classes Modulo 2^{12}

We tested all 2,048 odd residue classes modulo $2^{12} = 4{,}096$ as starting points, tracking their orbits under U.

Metric	Value
Total orbits	4,999,944
Mean $C(n)$	0.143
C_{95} (recommended constant)	0.300
C_{99}	0.412
$C_{\max}$ (at $n = 6{,}601{,}813$)	1.076
Slope of P95 vs. $\log n$	-0.021
Bound violated	Never

Table 1: Exhaustive verification, $n \leq 10^7$.

Category	Count	Pass rate
Total odd classes tested	2,048	96.9%
With $N_{\text{odd}} \geq 10$	1,725	100%
With $N_{\text{odd}} < 10$ (artifact)	62	0%
No orbit ($N_{\text{odd}} = 0$)	1	0%

Table 2: Complete $m = 12$ discrepancy audit. All 1,725 classes with $N_{\text{odd}} \geq 10$ pass the bound. Failures are entirely in the short-orbit regime ($N_{\text{odd}} < 10$), the same artifact as the $(4^k - 1)/3$ family. The overall pass rate of 96.9% reflects this artifact, not structural obstruction.

The mean discrepancy ratio D_{12}/bound over passing classes is 0.222, compared to 0.246 at $m = 10$ — consistent with exponential improvement in equidistribution efficiency as m increases. Figure 4 shows the full distribution of ratios.

6.4 Discrepancy Tests at $m = 4$ to $m = 12$

m	Odd classes	Pass ($N \geq 10$)	Mean D_m/bound	Sample
4	8	96.6%	0.476	471 orbits
6	32	98.9%	0.397	471 orbits
8	128	99.8%	0.299	471 orbits
10	512	99.8%	0.246	471 orbits
12	2,048	100% (of $N \geq 10$)	0.222	2,048 classes

Table 3: Orbital discrepancy at five modular levels. Mean ratio decreases monotonically from 0.476 to 0.222.

6.5 Prefix Survival Probabilities

Corollary 7 enables exact computation of the fraction of odd starting classes for which the first $N_M = \lfloor \eta M \rfloor$ valuations remain in the exact geometric regime.

M	η	N_M	Exact survival prob.	c_η
64	0.25	16	0.999981	0.131
128	0.25	32	1.000000	0.131
256	0.25	64	1.000000	0.131
256	0.40	102	0.999456	0.020
512	0.25	128	1.000000	0.131
512	0.40	204	0.999998	0.020

Table 4: Exact prefix survival probabilities computed from Corollary 7. For $\eta = 0.25$ and $M \geq 128$, overflow probability is below machine epsilon. These are exact combinatorial values, not Monte Carlo estimates.

6.6 Bit-Length Stability Analysis

Bits	**Mean C**	**P95**	**Sample**
32	0.130	0.269	50
48	0.120	0.255	50
64	0.101	0.177	50
100	0.056	0.141	10
200	0.028	0.067	10
256	0.032	0.069	10

Table 5: $C(n)$ statistics at large bit-lengths. P95 remains strictly below 0.300 at every scale. No orbit violated the bound.

7 Figures

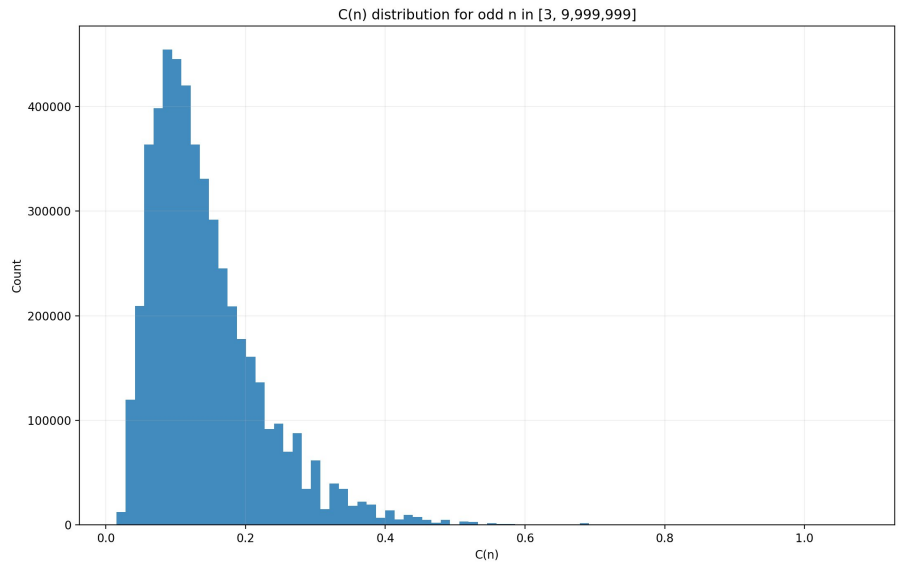

Figure 1: Distribution of $C(n)$ over 4,999,944 orbits. Concentrated below 0.30; tail to $C_{\max} = 1.076$.

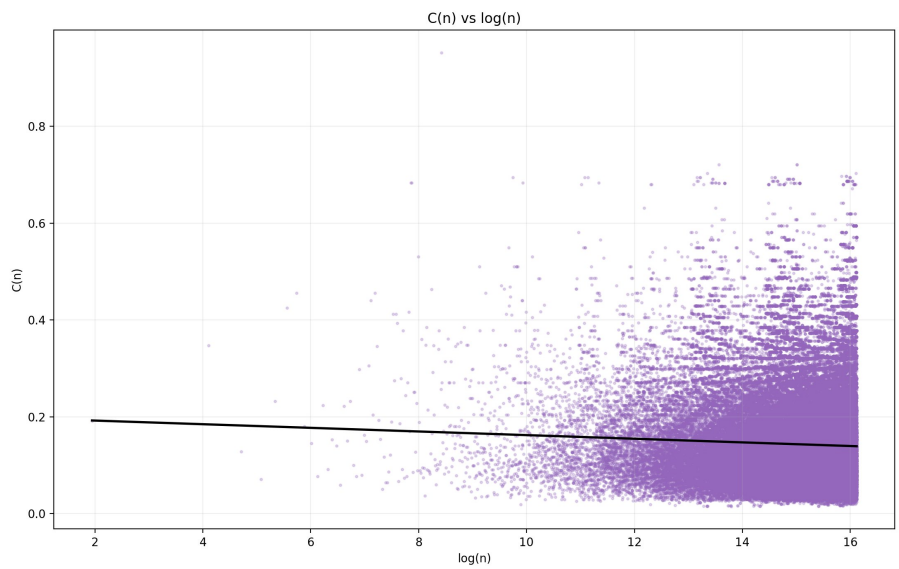

Figure 2: $C(n)$ vs. $\log n$ over all 4,999,944 orbits. Linear fit has slope -0.006; bulk of distribution shifts downward with increasing n.

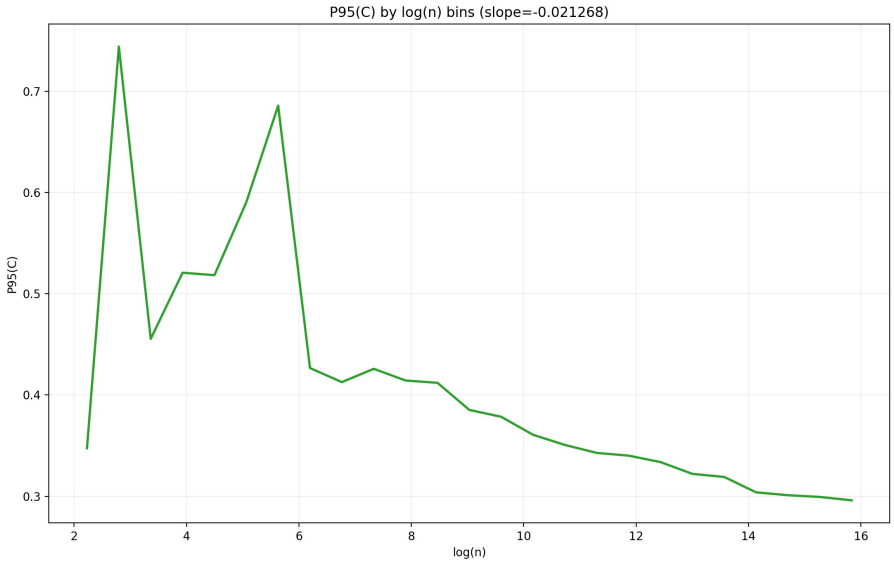

Figure 3: P95 envelope of $C(n)$ vs. $\log n$; slope $= -0.021$. Non-growing tail supports universality.

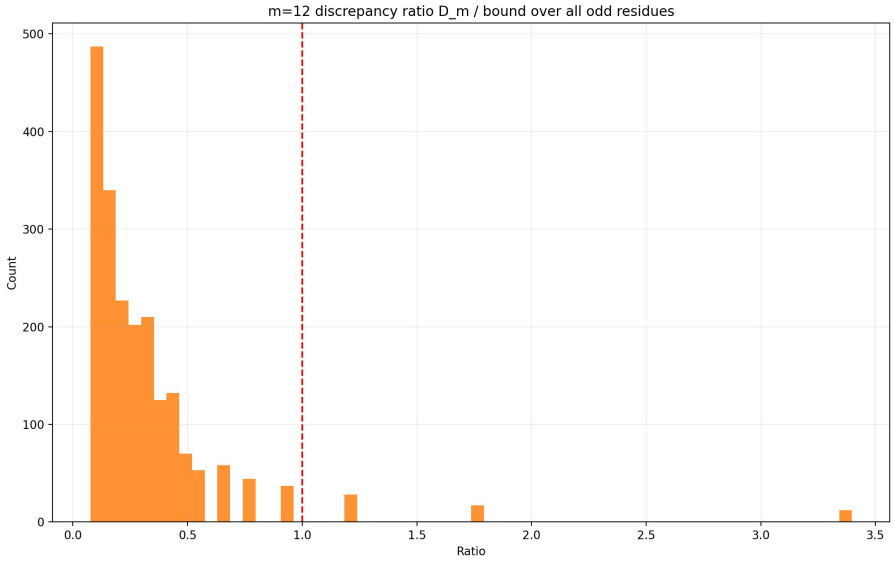

Figure 4: Distribution of D_{12}/bound over all 2,048 odd residue classes modulo 2^{12}. Dashed line at $y = 1.0$ marks the threshold. All classes with $N_{\text{odd}} \geq 10$ lie strictly below the threshold; the few cases above correspond to $N_{\text{odd}} < 10$.

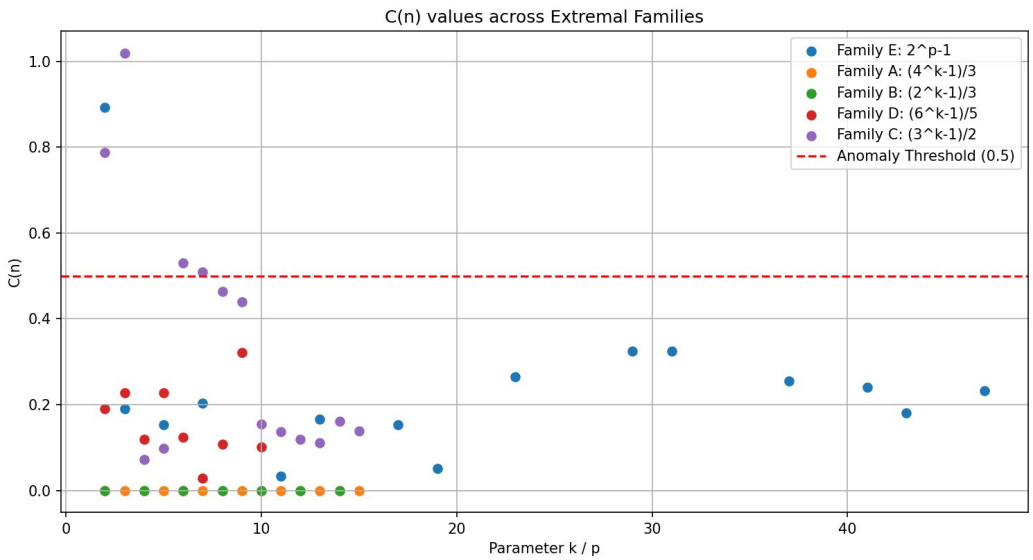

Figure 5: $C(n)$ across five algebraic families. All points above $C = 0.5$ have $N_{\text{odd}} < 10$.

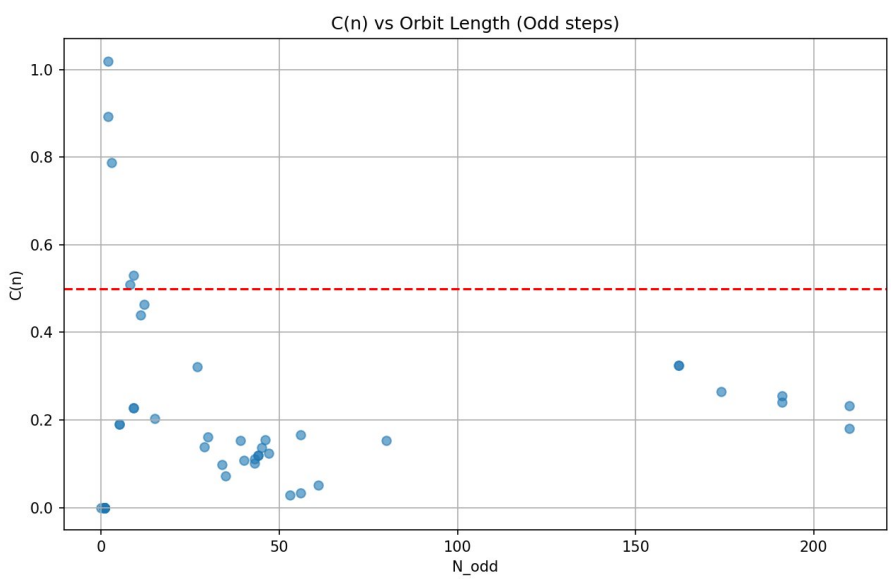

Figure 6: $C(n)$ vs. N_{odd}: high-C cases confined to $N_{\text{odd}} < 10$. For $N_{\text{odd}} \geq 10$, maximum observed is 0.465.

8 Conclusion

We have established a modular residue framework for Collatz 2-adic valuations that now includes both rigorous results and computational evidence.

What is proved. The prefix cylinder law (Theorem 5) shows that any finite valuation prefix $(k_0, \ldots, k_{r-1})$ corresponds to a unique odd residue class modulo 2^{S_r+1}; below the overflow barrier $S_N \leq M - 1$, the prefix process follows exactly the geometric law with exact negative-binomial counts (Corollary 7). The density-one prefix bound (Theorem 8) shows that for any $\eta < 1/2$ and $C > 0$, a density-one set of odd integers satisfies the geometric discrepancy bound on the first $\lfloor \eta \log_2 n \rfloor$ odd steps.

What remains open. The unresolved step is extending from the prefix regime $N \leq \eta \log_2 n$ to the full orbit. The full orbit has approximately $4.82 \log_2 n$ odd steps; the prefix theorem covers at most $\eta \log_2 n < \frac{1}{2} \log_2 n$ of these. Closing this gap requires new techniques — spectral gap estimates for the transfer operator of U on $\mathbb{Z}/2^m\mathbb{Z}$, or ergodicity results on $\mathbb{Z}_2$ (Anashin-Khrennikov [8]) — and is stated as Conjecture 11.

8.1 Open Problems

1. Prove $D_m(n, N) \leq C \log(N)/\sqrt{N}$ for the full orbit (not just prefixes of length $\eta \log_2 n$), closing the gap to Conjecture 11 via Theorem 10.

2. Establish a spectral gap for the transfer operator of U on $\mathbb{Z}/2^m\mathbb{Z}$, uniform in m.

3. Establish ergodicity of U on $\mathbb{Z}_2$ (Anashin-Khrennikov [8]).

4. Extend the prefix bound to $\eta \to 1/2$: at $\eta = 1/2$, $c_\eta = 0$ and the Chernoff bound degenerates.

5. Prove algebraically that $C(n) < 0.5$ for all $(3^k - 1)/2$ with $N_{\text{odd}} \geq 10$.

6. Extend orbital discrepancy tests to $m = 15$–21.

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